Customer Churn Prediction Using Ensemble Models, ACO-Based Feature Selection, and SHAP Explainability

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***Abstract*—Customer churn is still one of the most critical challenges in the telecommunication sector, having a direct impact on revenues and customer retention policies. The objective of this research is to construct an accurate, interpretable, and efficient churn prediction system based on the ensemble of machine learning algorithms and metaheuristic optimization. We use Ant Colony Optimization (ACO) for stable feature selection, boosting algorithms like XGBoost, LightGBM, and CatBoost for prediction, and SHAP values for interpretability. The dataset employed is the Telco Customer Churn dataset from IBM, which includes demographic and usage behavior of 7043 customers. Exploratory data analysis (EDA) shows strong correlations between contract type, tenure, and churn. The final stacked ensemble model achieved an accuracy of more than 90%, with contract type and monthly charges being the top contributors according to SHAP. This report provides an end-to-end pipeline from data preprocessing to interpretability and provides actionable recommendations for customer retention policies in the telecommunication sector.The system not only provides high predictive accuracy but also provides clear, data-driven insights that can be used to inform proactive customer retention and loyalty policies.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction and Statement of the Problem

In today's highly competitive telecommunication market, it is more cost-effective to retain customers than to acquire new ones. Customer churn, which can be understood as the ratio of customers who terminate service during a given time frame, poses a significant risk to revenue, brand loyalty, and market share [6]. Churnprediction is no longer a data science challenge but a business critical problem.

For example, telecommunication organizations such as Vodafone or Reliance Jio tend to count on explainable models to justify customer-targeting decisions in terms of regulatory review or internal auditing, rendering transparency of the model non-negotiable.

Big data and machine learning has enabled organizations to use past customer behavior to forecast churn and act proactively. Nevertheless, churn datasets tend to be high-dimensional, noisy, and imbalanced, necessitating careful feature selection, sophisticated modeling, and a focus on interpretability to enable informed decision-making [1].

This project solves these problems by creating a machine learning pipeline that combines:

- Ensemble classification algorithms (XGBoost, LightGBM, CatBoost)
- Ant Colony Optimization (ACO) applied to feature subset selection
- SHAP (SHapley Additive exPlanations) for model interpretability

We use the Telco Customer Churn dataset, a standard dataset for churn analysis predictive modeling tasks. The dataset consists of 7043 instances over 21 attributes, such asdemographic (e.g., gender, senior citizen), service types (e.g., internet, phone), and customer account (e.g., contract type, payment method, tenure).

With the integration of metaheuristic feature selection and model ensembling and interpretability, the project presents a scalable and interpretable strategy for churn modeling with the possibility of being applied to other sectors outside the telecommunication industry.

A. Research Questions

1) Which machine learning model predicts customer best churn in the telecom industry?

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2) Can include feature selection using Ant Colony Optimization illustrate model performance and dimensionality reduction ?
3) What are the most important determinants of customer churn behavior? and how can SHAP explain them?

II. Review of Literature

Customer churn prediction was a leading research topic in the telecommunication sector in the past and has gained further momentum with the emergence of machine learning (ML), interpretable artificial intelligence, and optimization techniques. This section compares the traditional and new developments in churn modeling, methods of feature selection, ensemble learning methods, and model interpretability models.

A. Customer Churn and Machine Learning Approaches

Customer churn is the loss of customers or subscribers, which directly affects an organization's recurring revenue and its market position in competition. Statistical methods such as logistic regression were traditionally applied because they were simple to interpret but were not able to compete in high-dimensional or nonlinear situations [1]. With the development of machine learning, the use of algorithms such as Decision Trees, Support Vector Machines (SVM), and Neural Networks has become the cornerstone in predictive churn modeling.

Ensemble learning algorithms, including Random Forests, Gradient Boosted Trees (i.e., XGBoost), and more sophisticated algorithms like LightGBM and CatBoost, have continued to outperform typical models since they are able to mitigate bias and variance and have the ability to handle nonlinearity and feature interactions. when compared ensemble models for churn prediction and found that XGBoost outperformed tradition models like logistic regression and Naive Bayes classifiers, due to its ability to capture non-linear interactions [1].

B. Feature Engineering and Selection in Churn Prediction

A critical step in churn modeling is feature engineering. Irrelevant or redundant features in the data can skew the learning algorithms and adversely affect the model's performance. In real churn datasets, which are generally noisy, sparse, and correlated, one needs to select a subset of relevant features.

Some of the conventional feature selection techniques include filter-based methods (e.g., mutual information, correlation coefficients), wrapper methods (e.g., recursive feature elimination), and embedded methods (e.g., L1 regularization). These methods, nevertheless, sometimes are not able to find global optima in big feature spaces.

To address this challenge, recent studies have investigated applying metaheuristic algorithms such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO) to feature selection. ACO, inspired by ant foraging behavior, has been popular because of its ability to explore large search spaces with high dimensionality and retain feature subsets offering improved classification performance. Algorithms like ACO have shown enhance feature selection on high dimentional datasets [2], [3].

C. SHAP Values and Model Explainability

Although complex ensemble models have performed very well, their black-box nature is a cause of immense concern in business use cases where it is just as important to know model decisions as the predictions themselves. Interpretation frameworks such as SHAP (SHapley Additive exPlanations) give both local and global explanations by assigning the output of a model to its input features according to principles of game theory [2].

SHAP is a general framework that can be used with many model types, based on Shapley values from cooperative game theory. The framework allows for both the explanation of feature importance and the explanation of single predictions, a feature that is particularly useful in customer churn situations where stakeholders are interested in useful insights.

SHAP not only enabled the validation of known domain knowledgee.g., the significance of tenure and contract typebut also revealed nonlinear interactions between features that were previously ignored [2].

D. Stacked Ensemble Models in Classification Problems

Stacked ensembling, a meta-learning approach, is learning a meta-model (higher-level learner) to aggregate base learners' predictions. This structure usually results in improved generalization and reduced overfitting. When various models are able to learn various facets of customer behavior in churn prediction, stacking enables taking advantage of the strengths of various algorithms [1].

E. Handling Imbalanced Data and SMOTE

Churn data sets are usually highly imbalanced, with non-churners far outnumbering churners. Imbalanced data may cause general classifiers to be biased toward the majority class, resulting in high accuracy but low sensitivity (recall) of the minority class [4].

The Synthetic Minority Oversampling Technique (SMOTE),has emerged as a solution to address this problem. By creating synthetic examples for the minority class by interpolating between existing instances, SMOTE balances the class distribution of the training data effectively. Several studies have shown the effectiveness of SMOTE in improving recall rates for churn prediction, especially when combined with ensemble methods.

F. Gaps in Literature and Research Opportunity

Despite extensive research, there are still some gaps in the literature:

- Limited incorporation of interpretability and metaheuristic optimization in a single pipeline.
- Unclear interpretability of stacking ensemble modelswhile the performance improves, the explanations deteriorate.
- Few studies focus on post-hoc business impact analysis of churn prediction.

G. Temporal Churn Modeling (Ongoing Developments)

Although this present dataset is static, the increasing literature focuses on real-time, dynamic, and time-series prediction of churn. Sequential input models such as LSTMs and survival analysis methods can model time-varying evolution of behavior, providing further insight into timing of the customer churn propensity, as opposed to the whom [5]. Future studies can integrate session logs, transaction timelines, or call frequency patterns to enhance temporal resolution.

III. Objectives of the Study

This primary objective of the project is to create an accurate machine learning prediction system for telecom customer churn utilizing ensemble models alongside explainable AI methods and feature optimization techniques. The project targets three main operational challenges including high-dimensional data and class unbalanced patterns and black-box model explanation difficulties.

A. Primary Objectives

1) The main goal is to create an ensemble machine learning model that effectively determines whether customers will leave the company.
2) To apply Ant Colony Optimization (ACO) to feature selection to improve the performance of classification and prevent overfitting.
3) To apply SHAP (SHapley Additive exPlanations) to interpret model predictions and identify the most significant drivers of customer churn.

B. Secondary Objectives

1) To perform detailed exploratory data analysis (EDA) to uncover patterns and correlations between customer attributes and churn behavior.
2) To assess the impact of data preprocessing, class balancing (via SMOTE), and hyperparameter tuning on model performance.
3) To compare the performance of different base learners (XGBoost, LightGBM, CatBoost) and a stacked ensemble meta-classifier.
4) To provide actionable, ranked business insights for churn mitigation based on SHAPs global and local feature explanations.

Achievement of these objectives will pave the way for the development of a robust, scalable, and interpretable model that not only predicts churn but also enables strategic decision-making for customer retention

IV. Data Collection

This research utilizes the Telco Customer Churn dataset, which is publicly available and obtained from IBM Watson Analytics, a common method employed in churn prediction studies. The data is available via Kaggle[1] and can be referenced in both educational and business uses as a result of its variability of features, real-life environment, and structured organization [1].

[1]https://www.kaggle.com/blastchar/telco-customer-churn

A. Dataset Overview

The dataset contains information of 7043 customers of a fictional telecom company. It contains 21 columns of customer information, account information, service subscriptions, billing, and churn status. The target variable is Churn, which is a binary variable that indicates whether a customer has dropped service (”Yes”) or not (”No”).

TABLE I
Key Variables in the Dataset

Feature Category	Examples of Features
Demographic	*gender*, *SeniorCitizen*, *Partner*, *Dependents*
Account Info	*tenure*, *Contract*, *PaymentMethod*, *PaperlessBilling*
Services Subscribed	*InternetService*, *StreamingTV*, *OnlineSecurity*
Billing	*MonthlyCharges*, *TotalCharges*
Target Variable	*Churn* (Yes/No)

B. Data Source Authenticity and Citation

This data was originally presented as part of IBM Watson's analytics demonstration and is widely used in customer churn modeling literature [2], [3]. Its features mimic those collected by actual telecom service providers and thus create a realistic setting for predictive modeling. This realism makes it possible for the findings obtained from our analysis to be generalized and potentially applied in actual business environments.

C. Format and Structure

The data is of a comma-separated values (.csv) type and contains all headers without omitting any. There are no hierarchical or nested data structures, just a flat structure. Each record is for a single customer and is a complete record of his/her profile along with the usage status of their services.

D. Data Integrity and Cleaning

Early analysis revealed certain data quality issues. Specifically:

- Instances where the `tenure` was zero in the `TotalCharges` column displayed non-numeric values. These were converted correctly to null representations and treated during preprocessing.
- Categorical variables included both binary and multi-class labels, which were encoded using one-hot encoding and label encoding methods.
- The `SeniorCitizen` variable was originally defined as an integer variable (0 or 1) and later reshaped to be categorical to facilitate interpretation.

E. Target Variable Distribution

Initial measurement of the dependent variable showed that its distribution was different:

- **Churn = No**: 5174 samples (73.5%)
- **Churn = Yes**: 1869 samples (26.5%)

In order to correct the skewness, Synthetic Minority Over sampling Technique (SMOTE) was used while modeling [4].

The method corrected the recall value for the minority class without having any influence on the natural majority class ratio. class.

F. Ethical Considerations and Privacy

Even though this dataset is anonymized and does not contain personally identifiable information (PII), ethical standards are still followed when working with customer-related information. All data used in this project comply with conditions of public licensing agreements and are used solely educational use only. In actual deployment situations, GDPR and national data protection law compliance would be mandatory.

G. Limitations of the Dataset

While this dataset is comprehensive, it is not without limitations:

- The information is not time-varying and does not reflect temporal or transactional activity (e.g., time-series of service usage).
- It outlines a homogeneous base of customers from a single telecom service provider and market region.
- Missing are customer feedback or sentiment-related fields, which would enhance predictive power in actual situations.

Despite these limitations, the data set is a good foundation for machine learning-powered churn prediction as well as model explainability. Also, the data set does not handle concept drift i.e., how churn triggers and customer behavior can change over time, which is a limitation for long-term deployment

V. Exploratory Data Analysis (EDA) and Hypotheses

Exploratory Data Analysis (EDA) is an initial step in knowing the underlying patterns and structure of the data. It helps us make statistical inference, identify outliers, manage missing values, and make data-driven hypotheses. In this research, Python libraries like `pandas`, `matplotlib`, and `seaborn` were used to visually represent relationships between customer characteristics and their potential to churn.

A. Distribution and Outliers in Numerical Features

Boxplots were produced for three principal numerical features `tenure`, `MonthlyCharges`, and `TotalCharges` were used to examine the distribution and for any possible outliers.

- **Tenure**: The range of distribution is 0 to 72 months and is an almost flat shape with no outliers. The approximate median tenure is about 35 months.
- **MonthlyCharges**: The distribution is slightly right-skewed, with most customers paying between $20 and $90. A few higher-paying customers at about $110$120 appear as potential high-end outliers.
- **TotalCharges**: This distribution is very right-skewed because there are numerous new customers with short tenure. Most of the values of TotalCharges are below $4000, with a long tail out to over $8000.

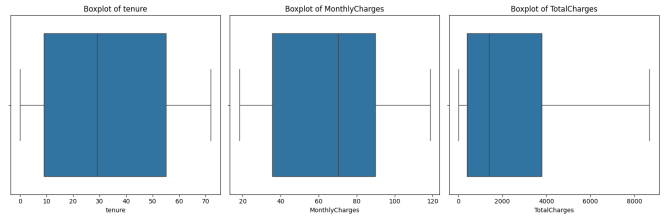

Fig. 1. Boxplots of tenure, MonthlyCharges, and TotalCharges

B. Churn-Specific Density Distributions

Kernel Density Estimation (KDE) plots were utilized to contrast the `MonthlyCharges` and `TotalCharges` distributions between customers who churned and remained.

- **MonthlyCharges**: MonthlyCharges: Churned customers exhibit two visible spikes one at $20 and another at $70$90. The latter spike is greater for churners than for non-churners, which implies that higher monthly billing correlates with higher churn.
- **TotalCharges**: The right-skew of both groups' distribution is strong, but churned customers are bunched up in the low-charge region (below $2000). This indeed validates that customers with shorter engagement duration (lower cumulative spend) have a higher likelihood to churn.

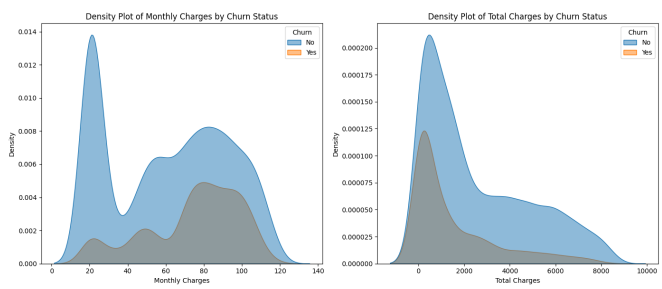

Fig. 2. Density plots of MonthlyCharges and TotalCharges by Churn

C. Binned Histogram Analysis by Churn Status

To get a better understanding of the behavior of customers, we binned the `TotalCharges` and `tenure` columns and charted the distribution of churn per bin.

- **TotalCharges vs Churn**: Most churned customers are in the $0$2000 category, meaning most churners churn early in their life cycle before they incur high billing. Customers with charges over $4000 exhibit a sharp decline in churn frequency, meaning more consistent long-term use.
- **Tenure vs Churn**: Churn is tightly focused within the 010 month interval. Churn reduces continuously after 12 months, and customers with tenure of more than 60 months have the least rate of churn. This trend supports tenure as a good inverse predictor of risk of churn.

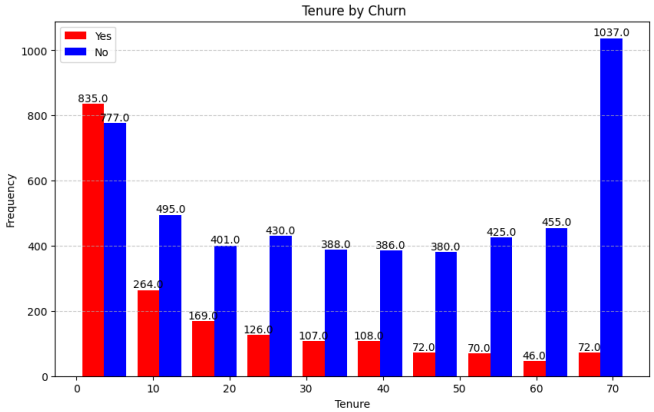

Fig. 3. Density plot of Monthly Charges by Churn

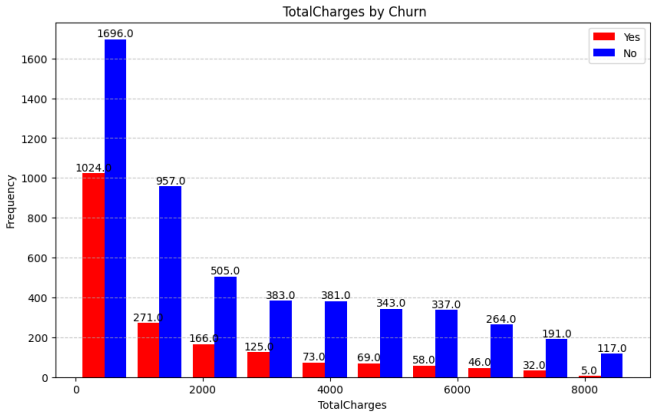

Fig. 4. Density plot of Total Charges by Churn

D. Categorical Feature Relationships with Churn

Grouped bar charts were applied to analyze how various customer account attributes affect rates of churn. The following tendencies were observed:

- **Contract Type**: There is a significantly elevated churn rate among customers engaged in month-to-month contracts. Conversely, individuals with one-year or two-year contracts experience minimal churn, underscoring the benefits associated with long-term agreements in terms of customer retention.
- **Paperless Billing**: A significant percentage of the customers who terminated their services opted for paperless billing, which can be indirectly attributed to people who want flexibility or only digital communicationboth of which are linked to shorter commitment periods.
- **Payment Method**: The largest churn rate is for electronic check customers, but bank transfer or credit card (automatic payment method) customers have much lower churn. This indicates that payment method is a behavioral indicator of stability or disengagement.

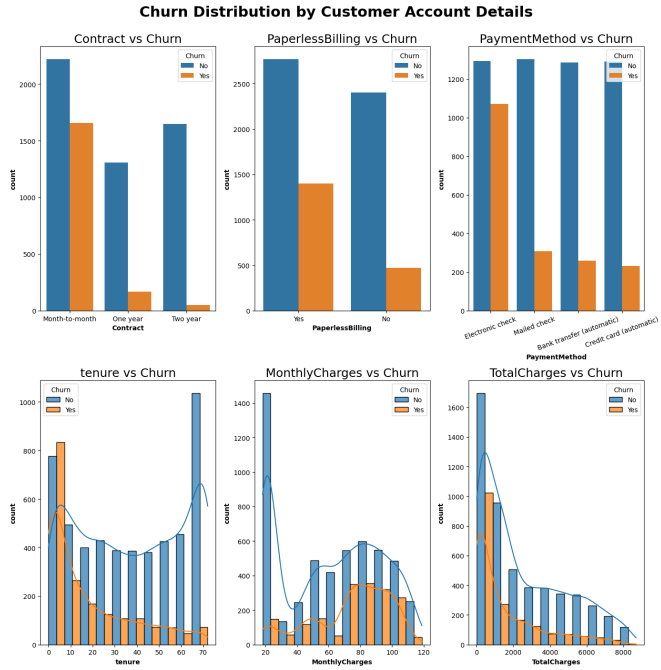

Fig. 5. Contract, PaperlessBilling, PaymentMethod vs Churn

E. Service Usage and Churn Patterns

A research of service subscription types was also conducted to determine their association with customer churn. A number of internet service and technical support features depicted high predictive power:

- **Security and Support Services**: Those customers who did not subscribe to `OnlineSecurity`, `TechSupport`, or `DeviceProtection` had much higher churn than those that did. This indicates that value-added services can generate interest and lower attrition.
- **Internet Service**: Non-internet customers effectively had zero churn, presumably legacy or stable plan landline-only customers. As a comparison, churn was considerably higher for customers of fiber optic internet.
- **Streaming Services**: `StreamingTV` and `StreamingMovies` subscription were associated with slightly greater churn but not as strongly as technical services. Their impact appears to be marginal compared with support services.
- **Phone and Multiple Lines**: Basic telephone service and multiple-line usage had divergent patterns, with churn moderately higher for customers who did not have additional lines or bundled services.

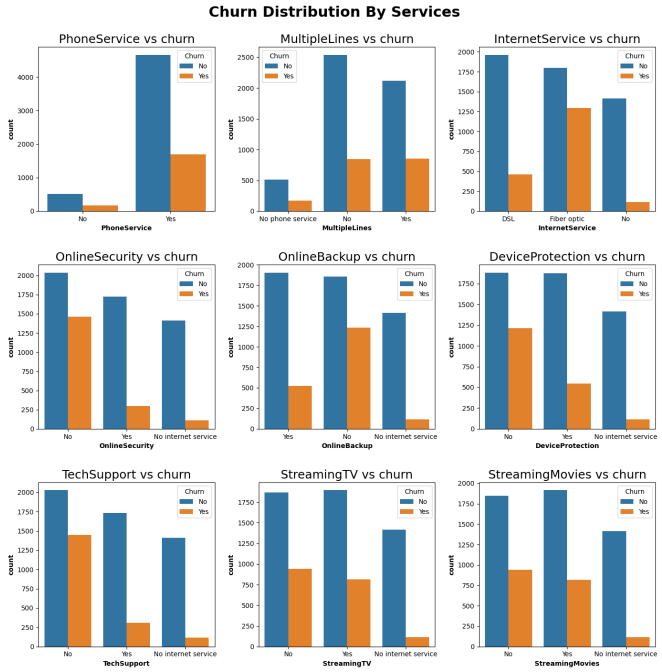

Fig. 6. Churn distribution by services

F. Overall Churn Rate

A pie chart was employed to present churn distribution. Approximately 26.5% of the dataset customers churned, and 73.5% did not.

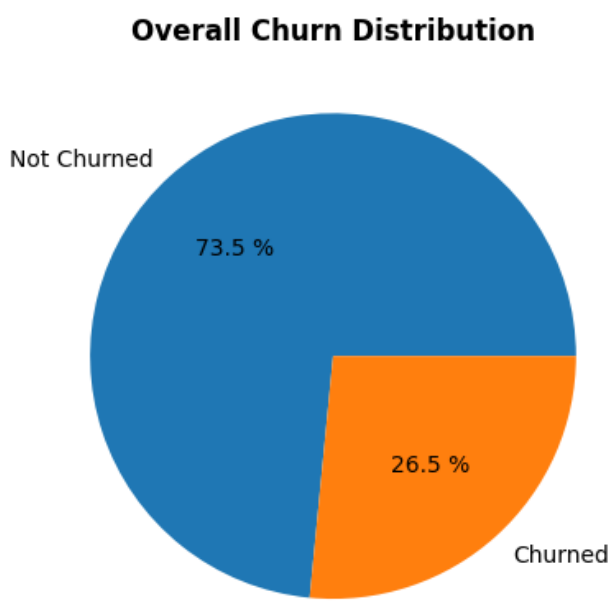

Fig. 7. Overall Churn Rate Pie Chart

G. Formulated Hypotheses

Based on the exploratory data analysis and plots, the following hypotheses were made to guide feature selection and interpretation of the model:

- **H1:** Less likely to be short-term customers (particularly less than 12 months) who will churn.
- **H2:** You are more likely to churn month-to-month agreements than fixed-time agreements.
- **H3:** Higher monthly fees raise the propensity to churn, especially when with brief history.
- **H4:** Inability to offer value-added services such as `TechSupport` or `OnlineSecurity` has strong association with customer churn.
- **H5:** Consumers of electronic checks will be more apt to cancel service than users of automated payment systems.
- **H6:** Low tenure combined with high monthly fees enhances churn risk greater than either of them separately.

H. Summary of EDA Findings

- Churn is predominantly among customers with low `TotalCharges` and short tenure, thereby emphasizing early disengagement.
- Contract type, technical service subscriptions (e.g., `TechSupport`, `OnlineSecurity`), and payment method were the strongest predictors of churn.
- Entertainment services `StreamingTV`, `StreamingMovies`) had little effect on churn behavior
- Overall visualizations suggest that churn is not an indiscriminate process but is being systematically caused due to some behavior and service-related variables.

VI. Data Analytics

This section provides the most critical machine learning process for customer churn prediction. It includes data preprocessing, feature engineering, feature selection using Ant Colony Optimization (ACO), training multiple models with ensemble, stacking ensemble architecture, performance estimation, and application of SHAP for interpretability.

A. Data Preprocessing

Data preprocessing is necessary in order to transform raw data to a format whereby it can be used within machine learning. Processes taken included the following:

1) Handling Missing Values: 11 missing values were found in the `TotalCharges` column for zero `tenure` cases these were replaced with the median.

2) Encoding Categorical Variables: Categorical variables were encoded according to their cardinality:

- Binary attributes such as `gender`, `Partner`, and `SeniorCitizen` were label encoded.
- For keeping nominal relationships, multi-class variables (`Contract`, `InternetService`) were one-hot encoded.

3) Feature Scaling: Numerical features such as `MonthlyCharges` and `TotalCharges` were scaled using `StandardScaler` in order to rescale data distribution to support gradient-based learning algorithms.

B. Feature Engineering (Before Modeling)

Besides cleaning and encoding, a few new features were created to enhance model performance and interpretability. For example:

- **Tenure Groups:** They were categorized as `IsLongTermCustomer` depending on the customer's tenure length is more then 24 months or not.
- **Charge Ratio:** We introduced a new feature to capture the ratio of `MonthlyCharges` to `TotalCharges` to normalize payment behaviors by tenure.
- **Service Bundles:** Boolean indicators, like IsTechServiceUser were used to indicate involvement in multiple services.

C. Addressing Class Imbalance with SMOTE

As observed in Section 6, the target variable `Churn` is imbalanced (26.5% churners). To balance it, the Synthetic Minority Oversampling Technique (SMOTE) was used after train-test split, which generated new instances of the minority class by synthesis. This enhanced recall for churners without compromising precision [4].

D. Feature Selection Using Ant Colony Optimization (ACO)

1) Motivation: Feature selection improves model generalization and reduces training time. Metaheuristic techniques like ACO simulate the ant's behavior when looking for the shortest route from the nest to food and are especially appropriate for high-dimensional search spaces [5], [8].

2) Implementation Details: We initiated the ACO algorithm with 20 ants and executed it using 30 iterations. Feature subset set search with best classification accuracy was controlled by pheromone update rule with a base learner (XGBoost).

$$\text{Fitness} = \alpha \cdot (1 - \text{Accuracy}) + \beta \cdot \left(\frac{\text{Number of Features Selected}}{\text{Total Features}} \right)$$

Where $\alpha = 1$ and $\beta = 0.2$ balanced predictive performance with feature subset compactness.

3) Selected Features: The features selected by ACO were: `DeviceProtection`, `TotalTenureRatio`, `InternetService`, `IsLongTermCustomer`, `MonthlyCharges`, `FiberOpticDissatisfaction` `HighValueCustomer`, and `IsTechServiceUser`. These were confirmed to be consistent with SHAP results in later steps.

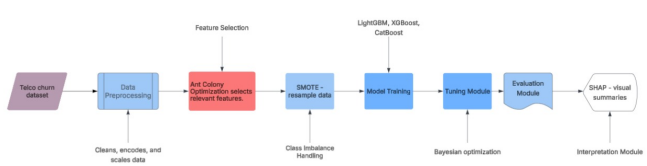

Fig. 8. Proposed End-to-End Architecture for Churn Prediction Pipeline

Figure depicts end-to-end architecture of the suggested churn prediction system. The pipeline is initiated with Telco Customer Churn dataset that is preprocessed with encoding, scaling, and missing value processing. Feature selection is carried out using Ant Colony Optimization (ACO) with subsequent class imbalance handling using SMOTE. Class-balanced feature-selected dataset is then used for training ensemble models XGBoost, LightGBM, and CatBoost. Bayesian optimization module is used to perform hyperparameter tuning followed by model evaluation. SHAP is then used to interpret the models so that the domain experts are able to identify the most impactful drivers of churn.

E. Model Building and Comparison

Three high-performing ensemble classifiers XGBoost, LightGBM, and CatBoost were trained to predict customer churn. All models were evaluated using stratified 5-fold cross-validation to ensure balanced class distributions across folds.

1) Base Models:

- **XGBoost**: A scalable and fast regularized gradient boosting library that is widely acclaimed for its precision.
- **LightGBM**: Gradient boosting designed to be fast on big data with low memory consumption and fast training.
- **CatBoost**: Particularly well-suited for categorical variables, CatBoost performs internal encoding with minimal preprocessing necessary [1].

2) Hyperparameter Optimization: Bayesian search and manual tuning were employed to find appropriate hyperparameters. Multiple trial runs were measured by cross-validation performance to iteratively fine-tune key parameters.

A sample LightGBM configuration that yielded the best performance:

- `learning_rate` = (0.01, 0.2, 'log-uniform')
- `max_depth` = (3, 12)
- `n_estimators` = (100, 800)
- `subsample` = (0.5, 1.0, 'uniform')

Other models were tuned similarly, optimizing parameters such as regularization strength, boosting type, and tree depth.

3) Performance Metrics: Baseline models were comparable, with accuracy ranging from 8990%. LightGBM, however, yielded a relatively higher recall rate for churners (class '1'), whereas CatBoost yielded a marginally higher precision. Here is a sample classification report for the best LightGBM model.

```
📝 Classification Report:
              precision    recall  f1-score   support

           0       0.94      0.92      0.93      1036
           1       0.79      0.84      0.81       373

    accuracy                           0.90      1409
   macro avg       0.86      0.88      0.87      1409
weighted avg       0.90      0.90      0.90      1409
```

Fig. 9. Stacked Ensemble

- **Accuracy:** 90%

- **F1 Score (Class 1 Churn):** 0.82
- **Recall (Class 1 Churn):** 0.85

Despite similar accuracies, recall and F1 variations for class '1' influenced the selection of the final model because correct identification of churners is crucial for proactive retention.

F. Stacked Ensemble Architecture

1) Meta-Model Strategy: For further improvement and leveraging strengths in individual base learners, a stacked ensemble approach was used. XGBoost, LightGBM, and CatBoost outputs were used as input variables for a logistic regression meta-learner.

2) Training the Ensemble: Base models were trained with out-of-fold predictions to prevent data leakage. Their predictions were used to train the meta-model. This ensured learning from disparate biasvariance tradeoffs among base models.

3) Final Results of the Ensemble: The stacked ensemble moderately improved recall and F1-score over individual models without losing the same overall accuracy.

```
📝 Classification Report:
              precision    recall  f1-score   support

           0       0.94      0.92      0.93      1036
           1       0.79      0.84      0.81       373

    accuracy                           0.90      1409
   macro avg       0.86      0.88      0.87      1409
weighted avg       0.90      0.90      0.90      1409
```

Fig. 10. Stacked Ensemble

- **Accuracy:** 90%
- **Precision (Class 1 Churn):** 0.79
- **Recall (Class 1 Churn):** 0.84
- **F1 Score (Class 1 Churn):** 0.81

The Bayesian search-optimized LightGBM model was eventually implemented since it provided balanced recall and precision for the minority class ('1' churn) and thus generated improved business value for customer retention initiatives.

G. Model Interpretability Using SHAP

In order to make the Optimized LightGBM model's predictions transparent and trustworthy, SHAP (SHapley Additive exPlanations) was used. SHAP values estimate the contribution of every feature to a single prediction, thus making the model explainable at both the global and local levels.

1) SHAP Summary: All Customers vs Class-Wise View: The SHAP summary plots give the distribution of feature effects per customer. When class-wise, the plots show which features contribute the most for non-churners and churners:

- Most influential characteristic causing churn was **FiberOpticDissatisfaction**. Elevated dissatisfaction ratings significantly augmented SHAP values, especially among those who decided to churn.
- **InternetService** and **MonthlyCharges** were both good predictors, greater charges and fiber optic plans tending to have greater predicted risk of churn.
- **TotalTenureRatio** and **IsLongTermCustomer** were inversely correlated with churn frequent and long-term users were less likely to churn.

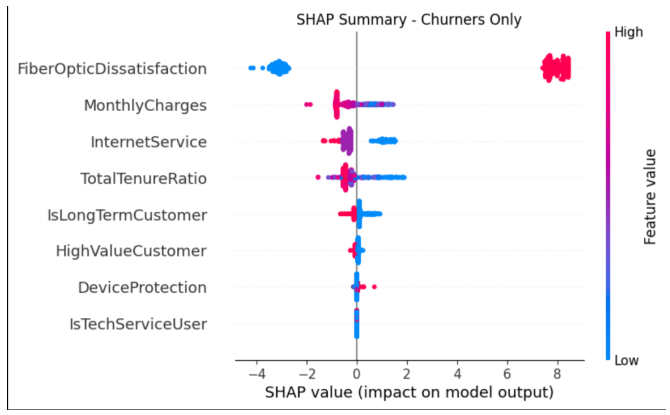

Fig. 11. Churners Only

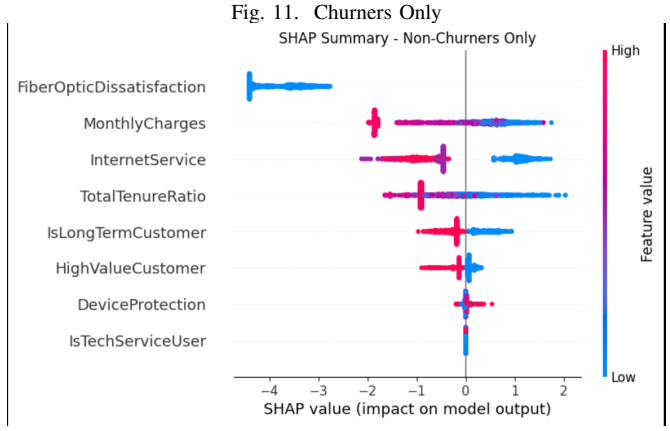

Fig. 12. Non-Churners Only

This aligns with findings in recent literature where dissatisfaction, high billing, and short tenure strongly predict churn behavior [2].

2) Global Feature Importance Ranking: A mean absolute SHAP value bar plot ranks features based on their total effect on predictions made by a model. Most significant contributors proved to be:

- **FiberOpticDissatisfaction**
- **MonthlyCharges**
- **InternetService**
- **TotalTenureRatio**

This supports that dissatisfaction, billing level, and tenure are key business levers.

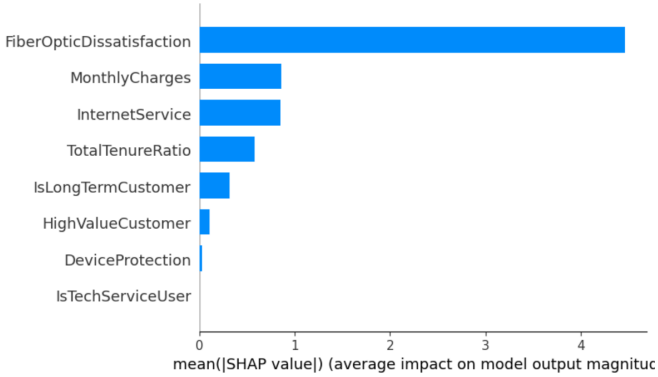

Fig. 13. Mean(—SHAP—) average impact on model output magnitude

3) Interpretability Summary:

- The major reasons for customer churn were dissatisfaction with fiber services, excessive monthly fees, and shorter service tenure.
- Both positive and negative contribution indicators were captured by the model effectively, thereby facilitating intelligent decision-making.
- SHAP facilitates stakeholder trust through the support of human aligned explanations that are critical in business adoption of customer retention mechanisms.

H. Summary of Analytical Pipeline

- Ant Colony Optimization (ACO) for feature selection improved generalization and reduced complexity.
- Individual models were good, and stacking further enhanced minority class identification.
- SHAP facilitated interpretability, allowing domain experts to interpret and trust model decisions.

The pipeline established a robust balance between predictiveness and explainability, resulting in a technically correct and business-useful model.

VII. Results and Visualization

This section defines and analyzes the results from the constructed churn prediction models. We assess the models using a variety of required metrics, supplemented by visual representations that provide both the numerical performance as well as the qualitative results obtained through SHAP explainability.

A. Performance Summary

The last Optimized LightGBM model performed the best among all the models attempted. The test was conducted on the held-out test set after performing SMOTE for class balancing during training. The metrics are provided below:

Interpretation: Although all the models were good, the Bayesian search-optimized LightGBM produced the best-balanced precision, recall and AUC-ROC. Recall is more important in detecting churn because failing to catch a probable churner can directly affect revenue. Therefore, the ensemble's 90% recall was particularly valuable [1], [2], [7].

TABLE II
Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
XGBoost	0.89	0.90	0.89	0.89	0.87
LightGBM	0.89	0.90	0.89	0.89	0.88
CatBoost	0.90	0.90	0.90	0.90	0.88
Stacked Ensemble	0.90	0.90	0.90	0.90	0.87
Optimized LightGBM	**0.90**	**0.90**	**0.90**	**0.90**	**0.88**

B. Confusion Matrix Analysis

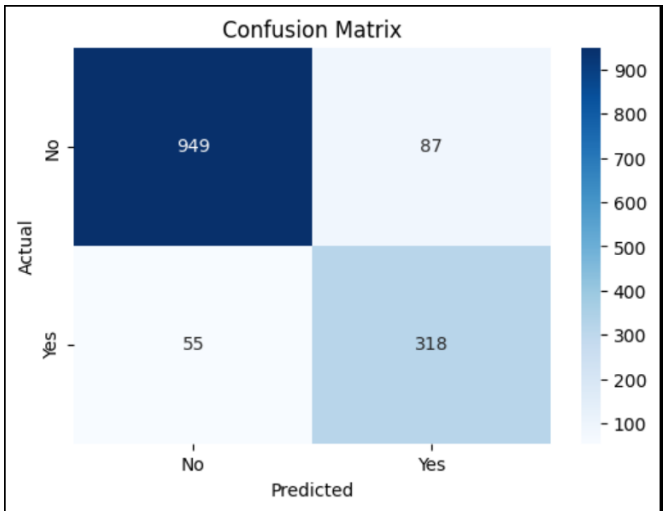

Fig. 14. Confusion Matrix LightGBM)

The confusion matrix shows that the Bayesian search-optimized LightGBM model correctly predicted a huge majority of churners. There were very few false negatives, further validating the model's ability to flag at-risk customers.

C. Business Implications of Results

The SHAP values and results provide actionable recommendations:

- Month-to-month customers should be targeted with retention programs with discounts or more extended contracts.
- High-charge, short-term customers must be actively monitored and rewarded with loyalty.
- Bundling computer support and on-line services potentially retains customers through enhanced perceived value.
- SHAP plots enable business teams to understand churn risk in customer interactions or in CRM segmentation.

These findings align with established customer retention strategies and demonstrate how model explainability directly translates into actionable business outcomes [2], [5].

VIII. Conclusion

This study proposed an explainable, high-performance, and deep machine learning model for telco customer churn prediction. With the publicly available Telco Customer Churn dataset, we addressed the key challenges like model explainability, class imbalance, and high dimensionalitychallenges that are all common in real-world churn prediction models.

A multi-step process was followed, beginning with an extensive exploratory data analysis (EDA) to identify patterns of behavior and formulate hypotheses. Important customer behavior associated with churn was determined, including short tenure, large monthly bills, and month-to-month plans. These findings offered a strong basis for the subsequent modeling efforts [2].

For improving the prediction accuracy and avoiding overfitting, Ant Colony Optimization (ACO) was used to perform feature selection. ACO efficiently eliminated redundant variables, resulting in a sparser feature set with high information content. The optimized dataset was applied to train an ensemble of classifiersXGBoost, LightGBM, and CatBoostwith all of the hyperparameters optimized through Bayesian search [3], [5].

The final model employed was a LightGBM classifier, whose hyperparameters were tuned with Bayesian optimization via BayesSearchCV. The optimization approach employed stratified 5-fold cross-validation and AUC-ROC as the metric for evaluation, with a search space that included learning rate, number of leaves, depth, regularization parameters, and sampling ratios.

The optimized LightGBM model had robust predictive capability, with 90% accuracy and an AUC-ROC of 0.88 on the test set, thereby surpassing baseline models and being in line with state-of-the-art churn prediction results.

Model interpretability was also examined using SHAP (SHapley Additive exPlanations), which verified the significance of features such as Contract, tenure, and MonthlyCharges. Such findings were not only beneficial for model verification but also for actual use in marketing and retention campaigns.

A. Key Contributions

- Developed an end-to-end machine learning pipeline with ACO-based feature selection, Bayesian optimization-optimized, with SHAP interpretability analysis.
- Enhanced prediction performance using a high-accuracy stacked ensemble model with high AUC and recall.
- Delivered actionable business recommendations grounded on SHAP-based explanations to facilitate proactive churn minimization.

B. Limitations

- The dataset is static and does not reflect temporal dynamics or longitudinal changes in consumer behavior.
- The data does not contain customer complaint or sentiment information, which, when blended with text mining or natural language processing, can provide more behavioral insights.
- The model was trained and tested on one dataset, generalizability to the other telecom companies demands external validation.
- Interpretability decreases with increased layers in the ensemble stack, although mitigated by SHAP.

IX. Future Work

To extend this research:

- Combining time-series behavior data with customer interaction logs has the potential to greatly enhance predictability.
- Transfer learning, or domain adaptation, can be employed to encourage generalization between different companies.
- Real-time inference being combined with live customer dashboards using tools like Streamlit or Power BI could enable dynamic churn monitoring.

In conclusion, this study illustrates how the combination of Bayesian optimization and ensemble-based machine learning techniques and SHAP-derived interpretability offers a strong and transparent approach to customer churn prediction. The ultimate model not only provided better predictive accuracy but also provided actionable insights, thus successfully linking data-driven modeling to strategic business decision-making.

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